

BRAC UNIVERSITY
CSE331 : Automata and Computability
Summer 2026

Duration: 30 minutes

Quiz 5

Total: 20 marks

Name:

ID:

0.5 Points

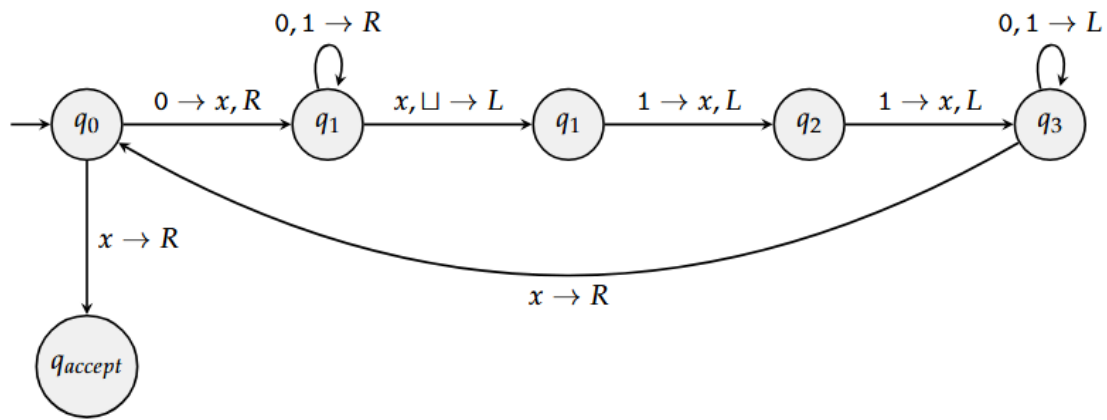
Section:

0.5 Points

Turing Machines

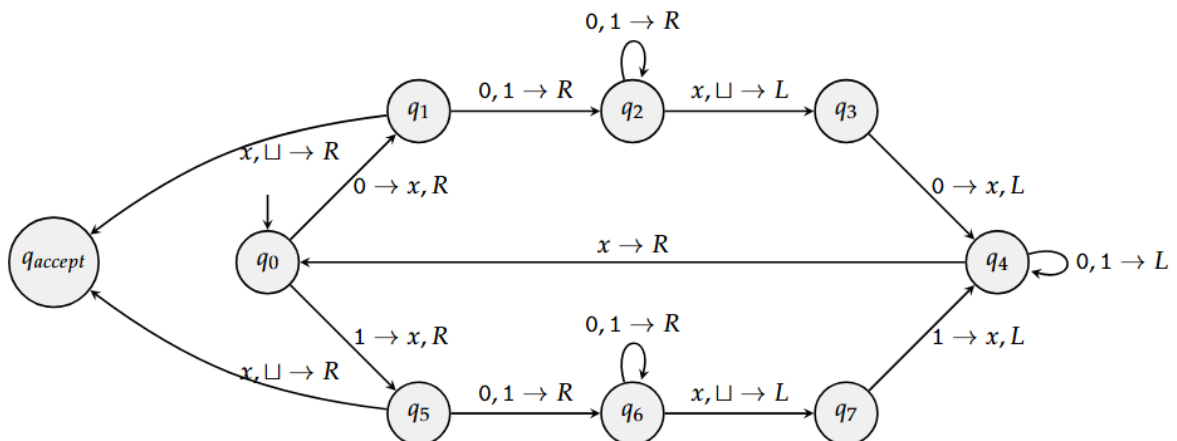
1. Let, $L_1 \rightarrow \{w = 0^n 1^m \mid n, m > 0 \text{ and } 2n = m\}$, where $\Sigma = \{0, 1\}$. Draw the state diagram of a TM that decides L_1 . (5 points)

Solution:



2. Let, $L_2 \rightarrow \{w \in \Sigma^* \mid w \text{ is an odd length palindrome}\}$, where $\Sigma = \{0, 1\}$. Draw the state diagram of a TM that decides L_2 . (6 points)

Solution:



3. Write the current state and current head position of a TM in a configuration 01q₅100. (2 points)

Answer: q₅, 2nd '1'.

4. Given that

$A_{DFA} \rightarrow \{\langle D, w \rangle \mid D \text{ is a DFA that accepts the string } w\}$ is decidable, prove that

$A_{CoRE} \rightarrow \{\langle R, w \rangle \mid R \text{ is a RE that does not generate the string } w\}$ is decidable. (6 points)

Solution:

Assume that M_{DFA} decides A_{DFA} .

We construct the following decider for A_{CoRE} :

$M_{CoRE} =$ "On input $\langle R, w \rangle$:

1. Convert RE R to an equivalent NFA N using Thompson's construction method.
2. Convert NFA N to an equivalent DFA D using subset construction method.
3. Simulate M_{DFA} on input $\langle D, w \rangle$.
4. If M_{DFA} accepts, reject; otherwise accept."

Since M_{DFA} is a decider, M_{CoRE} is also a decider.